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RESEARCH BRIEFING

The Effect of Social Networks on Resource Access and Business Start-ups

JAN INGE JENSSEN* and HAROLD F. KOENIG**

ABSTRACT *In previous research, entrepreneurs' social networks are expected to influence entrepreneurial success because they give access to resources. This is confirmed in this study. In most research it is also believed that network features such as strength of ties determine to a great extent what kind of resources the entrepreneur receives. Although this last assumption has important implications for entrepreneurship research and for practice, the validity of the assumption has not been put to the test in previous research. The analysis of data from an empirical study of entrepreneurs in Norway gives at best partial support for the assumption. Contrary to what should be expected, the results indicate that strong ties are important channels for information and that weak ties give access to finance.*

1. Introduction

Previous research reviewed in Jenssen (1999) has shown that social networks have a significant effect on entrepreneurship. In most of this research it is assumed that network features such as the strength of relationships determine the type of resources provided through the network. For example, it is assumed that weak ties are important because they provide the entrepreneur with information resources (Granovetter, 1973), and that strong ties give access to other kinds of resources (Krackhardt, 1992). However, in previous empirical research, except in Foss (1994) and Jenssen (1999), the analysis has been of the direct relationship between social networks and entrepreneurship, and not the relationship between the social network and entrepreneurship through resources. The questions to be addressed in this article are how the strength of ties affects resource access, and how resource access influences the chance for success of the business start-up. An answer to these questions is vital for how future research is conducted and also for what recommendations are given to prospective entrepreneurs.

Social networks are patterns of lasting social relationships between people (Greve & Foss,

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1990), and in this study the networks are investigated from the entrepreneur's point of view (ego-centred). Social networks are an important part of an entrepreneur's social capital (Coleman, 1988) which is a quality created between people that increases the return of a person's human capital such as intelligence and education (Burt, 1997). The term entrepreneurship may cover the creation of new enterprises, innovation, and even the management of small businesses (Mosted, 1991). In this study, the concept is defined as the start-up of new businesses.

2. Theory and Hypothesis

The social network approach and resource dependency theory are closely related. In resource dependency theory it is argued that organizations are dependent upon the exchange of resources (Pfeffer & Salancik, 1978; Thompson, 1967). The network approach to entrepreneurship is based upon the premise that a network is important because it provides the entrepreneur with access to necessary resources. For this reason, the arguments given in previous research concern both how social networks give access to resources and how the network affects the chance for entrepreneurial success. However, Jenssen (1999) has found that in most studies, resources have not been included as an intervening variable. In other words, the arguments concerning the relationship between social networks and resources are not based upon empirical knowledge.

Kanter (1983) categorized resources into three broad areas: (1) information; (2) motivational resources; (3) material resources. This categorization seems to cover most relevant types of resources discussed in the literature and it will be used to examine the resources required by an entrepreneur for the following hypotheses.

Access to information is typically considered to be critical for entrepreneurs. For example, Cooper *et al.* (1995, p. 108) argue that the process of venture formation might "be viewed as a process of learning, of overcoming the liabilities of newness through information acquisition". An especially important source of information is the weak ties (i.e. acquaintances and friends of friends) of the entrepreneur (Aldrich & Zimmer, 1986; Cooper *et al.*, 1995; Granovetter, 1973).

Burt's (1992) discussion of Granovetter's thoughts might indicate that it is the type of information that goes through the weak ties that is of importance. Weak ties do not necessarily give more information, but they provide non-redundant information which is of higher value.

However, many researchers who refer to Granovetter's (1973) work seem to believe that information in general (not only more valuable information) flows through weak ties more often than through strong ties. For example, Dubini and Aldrich (1991) argue that if you have a strong relationship to another person, you will also know the friends of the other person. Both parties in such a dyad will know much of what the other party knows. Therefore, the flow of information between two persons may probably decline with the strength of the relationship. This implies that a network consisting of strong ties will usually be very conformed and that information most often flows through weak ties. Greve and Gattiker (1994) seem to take this position, they state that information is largely obtained through weak ties.

Based on these arguments we have formulated the following hypothesis:

Hypothesis 1: The entrepreneur's weak ties more often give access to information resources than strong ties do.

Although several researchers assert that weak ties are important for receiving the necessary information, many have also pointed to the significance of other network features for access to resources other than information (Dubini & Aldrich, 1991; Greve & Gattiker, 1994;

Johannisson, 1988). For example it is assumed that a network consisting of strong ties gives the mental and social support that is necessary to promote entrepreneurial action (Johannisson, 1988).

Founded on arguments in the social network literature (Marwell *et al.*, 1988) it has been accepted that a close group of friends around the entrepreneur is important for generating action (Aldrich & Zimmer, 1986). Johannisson (1988) points out that this helps the entrepreneur to mobilize cognitive and emotive resources, self-confidence and the justification of choices already made by the entrepreneur.

Based on this we assume that there is a positive effect of strength of ties on access to motivation. However, the entrepreneur also needs other resources such as finance and practical help. It might be argued that it is hard to get such resources through acquaintances. We expect that it is necessary for an individual to have a strong relationship with an entrepreneur before the former is willing to 'give' such resources. In other words, we expect that there is a positive relation between the strength of the tie and access to motivation and finances. According to this, we can formulate the following hypothesis.

Hypothesis 2: The entrepreneur's strong ties more often give access to motivation and finances than weak ties do.

Weak and strong ties are assumed to have a positive effect on business start-ups because they give access to information, motivation, and finances (Granovetter, 1973; Aldrich & Zimmer, 1986; Krackhardt, 1992; Jenssen, 1999). This leads to the final hypothesis.

Hypothesis 3: The higher the number of accessible sources of information, motivation, and finances, the higher the chance for business start-up success.

3. Methodology

This study is a survey in which 100 persons who had tried to start a new business have been interviewed. The interviews were done by a combination of mail and telephone. The questions raised in this study are causal because we are interested in knowing how differences in the social network affect access to different kind of resources. Several adjustments to the survey design have been made in order to increase the reliability of conclusions (e.g. measuring possible threats, asking retrospective questions based on a classification of entrepreneurial phases, making sure that the selection of respondents included non-successful entrepreneurs).

The entrepreneurs were selected from individuals who had made contact with a centre for entrepreneurial training in Norway. The response rate was 64.5%. The selection was drawn randomly from a group of individuals who all had a desire to start a new business and had sought advice from the centre of entrepreneurial training on how to go about it. In the archive of entrepreneurs at the training centre there were relatively few failures. Therefore, we used a selection procedure to assure that we got enough failures in our sample. The response rate among failures was 60.3%.

In the analysis of data quality the outliers have been investigated and the variables have been checked for multicollinearity. None of these problems have caused difficulty. Also the distribution properties of the data have been tested. In order to attain approximate, normally distributed variables, we transformed most of the variables by their natural logarithms.

The independent variable in this study is social networks, which is defined as patterns of lasting social relationships between people. The networks are investigated from the entrepreneur's point of view (ego-centred).

Table 1. The effect of weak and strong ties on access to information resources

Independent variables	Dependent variable: information resources	
	Beta	Sig T
No. of weak ties	0.54	0.000
No. of strong ties	0.41	0.000

Adjusted $R^2 = 0.53$; significant $F = 0.000$; $N = 99$.

Degree of friendship is applied as the measure of tie strength. This is according to Krackhardt (1992), the single best measurement of strength. A tie is considered to be weak if the entrepreneur describes the contact as an acquaintance or loose acquaintance. It is regarded as strong if the entrepreneur describes the contact as a friend or close friend.

The intermediate variable, resources, is defined as the number of resources related to the business start-up that the entrepreneur receives. The type of resources that are needed for entrepreneurship is poorly discussed in the literature (Jenssen, 1992). Examples that are applied in studies are finance, information, expertise, advice on handling bureaucracy, advice on accounting, advice on technology, social support, encouragement, access to market/customers, suppliers, production resources, labour, etc. (Foss, 1994; Greve & Foss, 1990; Jenssen, 1999). As mentioned previously, Kanter's (1983) categorization of resources into information, motivation, and material resources will be used.

The dependent variable is the business start-up. Entrepreneurs who actually started a business are regarded as successes and those who failed to establish a business are regarded as failures. In order to be viewed as 'established', the company has been formally registered and it has had sales revenues.

4. Results and Analysis

The intervening variable, the number of resources the entrepreneur receives, are divided into three categories (Kanter, 1983); information, finances, and motivation. Information is measured by a six item scale (expertise, advice, information, information on customers, suppliers, and financial sources). A factor analysis of these items resulted in a one factor solution indicating that the items are unidimensional, and the Cronbach's alpha was 0.86 indicating the scale is reliable.

Financial connections and motivational connections are measured as the number of contacts from whom the entrepreneur received financial funding and motivation for the business start-up, respectively. Both of these items were directly worded and would be unlikely to be misinterpreted or misunderstood.

Variable 1, which is motivation and variable 4, which is finance are applied as separate variables in the testing of the hypotheses. Hypothesis 1 is tested in Table 1.

Table 1 shows that the beta-value is higher for the relationship between weak ties and information resources than between strong ties and information resources (beta = 0.54 versus 0.41). We know that the beta-value in regressions allows for a direct comparison between the independent variables as to their relative explanatory power (Hair *et al.*, 1992). This means that the explanatory power of weak ties on access to information resources is stronger than the explanatory power of strong ties on access to information resources. In other words, weak ties seem to give better access to information resources than strong ties do. This is consistent with

Table 2. The effect of weak and strong ties on access to information resources for the entrepreneurs who succeeded in the effort to start a new business

Independent variables	Dependent variable: information resources	
	Beta	Sig T
No. of weak ties	0.55	0.000
No. of strong ties	0.43	0.000

Adjusted $R^2 = 0.57$; significant $F = 0.000$; $N = 58$.

hypothesis 2. However, the regression confirms that both weak and strong ties give access to information resources at a significant level ($p < 0.001$ for both relationships). In other words, information seems to go through both weak and strong ties at a fairly high rate and the differences are not as great as some of the social network theorists seem to assume. Also, it has to be noted that a closer look at the regression results indicates a relatively large overlap between the confidence intervals (significance = 0.95, confidence intervals = 0.52–0.88 and 0.35–0.71). Therefore, the results may only be viewed as an indication of how strength of ties and information resources relate.

The general argument proposed in hypothesis 1 was that information flows through weak ties with higher frequency than through strong ties because both ends of a strong relationship will know what the other knows. However, Burt’s (1992) point is that lower redundancy in social networks increases the chance of entrepreneurial success because it gives access to more valuable information, i.e. non-redundant information. He also argues that weak ties and redundancy are correlated positively. If this line of argument is correct, the successful entrepreneurs will receive more information through weak ties than the non-successful entrepreneurs. Most likely there will also be a larger difference between the amounts of information received through weak rather than through strong ties for entrepreneurs with success, than for entrepreneurs that fail.

In Table 2 this argument is tested. Only the entrepreneurs who created new businesses (successful entrepreneurs) are included.

This analysis does not change the main picture of the results found in Table 1. The difference between the beta values is a little less than the difference revealed in Table 1. As for the regression results shown in Table 1, there is still a relatively large overlap between the confidence intervals in the regression shown in Table 2 where the relationship between weak ties and strong ties on access to information resources is examined. The confidence intervals are 0.45–0.88 and 0.30–0.71.

In hypothesis 2, it is hypothesized that motivation and finances more often flow through strong ties than through weak ones. Hypothesis 2 is tested in Table 3.

As expected, the beta value is higher for the relationship between motivation and strong ties than it is for weak ties (beta value = 0.46 vs. 0.11). It indicates that motivation more often goes through strong ties than through weak ties. This is as expected in hypothesis 2. However, there is a minor overlap in the confidence intervals (–0.7–0.28 and 0.28–0.63).

Finances go through weak and strong ties at about the same rate (beta = 0.28 versus 0.30). This result is not as expected in hypothesis 2. There might be several reasons for this. Finance might be acquired both through friends and through a regular market transaction. In the financial market it will not always be necessary to have a strong relationship in order to obtain money for a business start-up.

Table 3. The effects of strong versus weak ties on access to motivation and finances

Independent variables	Motivation ^a		Finance ^b	
	Beta	Sig T	Beta	Sig T
No. of weak ties	0.11	0.230	0.28	0.003
No. of strong ties	0.46	0.000	0.30	0.002

^aAdjusted $R^2 = 0.23$; significant $F = 0.000$; $N = 97$.

^bAdjusted $R^2 = 0.18$; significant $F = 0.000$; $N = 97$.

Table 4. The effect of resources and ties on the chance for business start-ups

Variables	B	Wald	Sig
Information resources	1.00	3.49	0.06
Motivation	− 0.37	0.59	0.44
Finance	3.48	18.31	0.00
No. of weak ties	0.58	0.00	0.98
No. of strong ties	0.54	0.01	0.94

R^2 : Cox & Snell = 0.38, Nagelkerke = 0.51; Sign = 0.000; 2 log likelihood = 87.26.

Prediction percentage correct: overall = 79.0; successes = 78.0; failures = 80.5.

An important question that we cannot answer through our study is whether the results would have been different for other material resources than finances. However, also other material resources will be obtained through ordinary market transactions and through friends. Therefore, we assume that other material resources will flow through both weak and strong relationships.

In sum, the results shown in Table 3 indicate that strong ties are more important for access to motivation than weak ties. This is as expected in hypothesis 2. However, for access to finance the difference between strong and weak ties is small.

In hypothesis 3 we hypothesized that access to information, motivation, and finances would influence the chance for business start-ups. The test of this hypothesis is displayed in Table 4. In the discussion earlier we assumed that the network affected business start-ups indirectly through resources. Tables 1–3 show that the number of ties influences resource access. In order to inspect whether there is a direct effect between the network and the chance for a business start-up the numbers of weak and strong ties are included in the logistic regression presented in Table 4.

Table 4 shows that it is the resource variables that are important for the chance for business start-ups. The network variables do not have any direct effect on the chance for business start-ups. This is in accordance with our expectations. However, the number of motivational resources does not seem to be important (for the chance for business start-ups). This is not in accordance with hypothesis 3.

5. Conclusions and Discussion

Before moving into the conclusions, we will identify areas that should be improved in future research. First, our sample is not large, and it only includes respondents from one country that sought assistance from a single centre for entrepreneurial training. Second, financial and

motivational connections are only measured as single items. However, Peter and Churchill (1986) point out that the validity of a construct is not only decided through reliability, but also via content validity. Although the two variables are directly worded thereby creating content validity, and single items are often used in behavioural research (Lievens & Moenaert, 2000), we still can not assess the amount of measurement error in each item. The next section begins to draw conclusions from the results.

The literature concerning social networks and entrepreneurship assumes that social networks affect entrepreneurship through the resources they give access to. This is confirmed in this study. However, our results indicate that the relation between social networks features and the resources provided through the networks is different from what the literature suggests. This observation is important. Wrong assumptions regarding the relation between social networks and resource access might lead to an incorrect inference of the influence of social networks on entrepreneurship and it might lead to inefficient advice on how the entrepreneurs should build their social networks in order to succeed.

Weak ties give access to information resources more often than other resources, and strong ties more often give access to motivation. This is in accordance with our expectations. However, the differences are not large and for finances there is almost no difference between weak and strong ties. These results are unexpected and seem to be different from the assumptions made in much of the literature on social network and entrepreneurship.

The relatively high and significant flow of information through strong ties raises doubt about the assumption that the entrepreneurs primarily rely on weak ties for access to information (see for example, Granovetter, 1973; and later Aldrich *et al.*, 1986, Cooper *et al.*, 1995; Dubini & Aldrich, 1991). Also, it raises doubt about Burt's (1992) argument that information received through weak ties (and non-redundant ties) is of more value for business start-ups than information received through strong ties. This is a tentative conclusion that is based on two arguments.

First, weak ties do not seem to be more important for access to information resources for the entrepreneurs that actually started a business than for those who failed. Second, when entrepreneurs are asked to name the persons from whom they received information they will probably remember the persons from whom they received valuable information.

An important question is why strong ties seem to be important for information. We will suggest three possible answers to this question. As for finances, some information might be regarded as so valuable that it will only be offered to friends. Also, some information might require extensive communication over time in order to be transferred from one individual to another. Such communication will probably most often be between friends. Finally, although the flow of information might decline with the degree of strength the flow of information will probably not decline to zero. Now and then one of the parties in a close relationship will provide the other party with valuable information.

The discussed results of our study raise questions that need to be addressed in future research. However, the results also have implications for entrepreneurial practice. Since information, motivation and finances flow through both weak and strong ties at fairly high rates, entrepreneurs should search for resources through their contacts regardless of the strength of the relationships.

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